THE SEPARABLE-FIRST SUBEXTENSION OF A NORMAL ALGEBRAIC EXTENSION IS NORMAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The lemma needs only the existence half of lemma-separable-first, whose uniqueness half was the reason the corresponding card was declined.

Editorial note. The label lemma-separable-first-normal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-separable-first (source lemma: E/E_sep/F with E_sep the elements separable over F; its card was declined on the uniqueness half, the existence half used here is carried), lemma-normally-generated (delivered card L48362: generators whose minimal polynomials split completely give a normal extension), E/F normal puts every root of an irreducible over F with a root in E into E

Source reference. Original reference 030K: sources/stacks-fields/pinned-original-source.tex, source lines 1921–1921. Prerequisite pointer invoked at 2081; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be a normal algebraic field extension. Then the subextension $E/E_{sep}/F$ of Lemma 030K is normal.*

Proof. If the characteristic is zero, then $E_{sep} = E$, and the result is clear. If the characteristic is $p > 0$, then E_{sep} is the set of elements of E which are separable over F. Then if $\alpha \in E_{sep}$ has minimal polynomial P write $P = c(x - \alpha)(x - \alpha_2) \ldots (x - \alpha_d)$ with $\alpha_2, \ldots, \alpha_d \in E$. Since P is a separable polynomial and since α_i is a root of P, we conclude $\alpha_i \in E_{sep}$ as desired. □

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